

EXECUTIVE SUMMARY

This report features in the EIT Digital policy perspective report series and is the result of a combined effort from five EIT KICs (EIT Manufacturing, EIT Urban Mobility, EIT Health, EIT Climate-KIC, and EIT Digital as coordinator). The EIT Knowledge and Innovation Communities represent the largest innovation ecosystem in Europe with more than 1,500 organisations from business, research, innovation, and higher education from all across Europe. By tapping into the vast innovation and application knowledge, the report identifies both general and sector specific concerns and opportunities for the further deployment of AI in Europe.

The report provides business and policy decision makers with a scenario-based impact assessment instrument. The scenarios off-set generic across the board application of policies and regulation against domain specific policies and regulation; and on the other hand, the level of regulation varying from soft to firm.

Next to the generic concerns and opportunities shared by all application domains, the report also identifies the key concerns and opportunities regarding AI in the domains of manufacturing, urban mobility, health, and climate and energy. This together with the core policy levels available to mitigate the concerns and embark on the opportunities.

Next to providing a pragmatic instrument for AI policy development, the main recommendations are:

- To ensure effective policy in the area of AI it is necessary to take context (sectors of application) into account.
- Policies regarding application of AI on personal data should be allowed to differ from policies regarding application of AI on machine data, especially in certain application sectors.
- General regulation or policy measures can be considered in relation to algorithm transparency and explainability.
- Regulation should be adaptable and flexible, whilst minimising and mitigating risks and ensuring human rights and European values.



INTRODUCTION

While a clear cut definition of Artificial Intelligence (AI) would be the building block for its regulatory and governance framework, there is not yet a widely accepted definition of what AI is (Buiten, 2019; Scherer, 2016). Definitions focussing on intelligence are often circular in that defining what level of intelligence is needed to qualify as 'artificial intelligence' remains subjective and situational¹. Pragmatic ostensive definitions simply group under the AI labels a wide array of technologies, applications, and uses (i.e., robots, cobots, machine learning and deep learning, computer visions, speech recognition, diagnostic systems, autonomous manufacturing machines, self-driving cars, and dozens more). The European Commission, for instance, in its 2018 Communication (European Commission, 2018, p. 2) defines AI as referring to: "systems that display intelligent behaviour by analysing their environment and taking actions – with some degree of autonomy – to achieve specific goals. AI-based systems can be purely software-based, acting in the virtual world (e.g. voice assistants, image analysis software, search engines, speech and face recognition systems) or AI can be embedded in hardware devices (e.g. advanced robots, autonomous cars, drones or Internet of Things applications)". Broad AI definitions cover several technologies, including machine learning (algorithms whose performance improve as they are exposed to more data over time), deep learning, predictive analytics, computer vision and natural language processing (see for instance Annoni et al., 2018). Following Buiten (2019), in this report we focus on algorithms and the data they use, for two reasons. First, algorithms and the underlying data is what most, if not all, the technologies and applications grouped under AI have in common. Second, they are at the same time the source of the new opportunities and risks that are currently the object of ethical and policy debates.

The European Commission expects AI to significantly improve the

lives of EU citizens and bring major benefits to society and economy through better healthcare, more efficient public administration, safer transport, a more competitive industry, and sustainable farming (European Commission, 2018b). Various AI applications are increasingly utilised in a diverse ranges of domains, including monitoring traffic congestion, employee hiring, metering smart energy grids, performing credit checks, assessing recidivism risks when deciding sentences, and many more (Teich, 2019; Teich & Tiras Research, 2018). They can guide decisions that up until recently only extensively trained humans could perform. This has allowed for remarkable improvements such as the ability to analyse medical images in radiology (Dreyer & Allen 2018) and to detect potentially cancerous cells (Al-shamasneh & Obaidellah, 2017) or predict where and when the next big earthquake may strike (Fuller & Metz, 2018). The potential for AI to overcome limitations of humans when dealing with computationally intensive tasks and augment intellectual and perhaps even creative capabilities, opens up new application domains, with impacts on productivity and performance (Dwivedi et al, 2019). While the applications are already in place, a recent EIT Health Report envisages more in healthcare in the near future, such as remote monitoring, AI-powered alerting systems or virtual assistants, giving patients increasing ownership of their care (EIT Health & McKinsey, 2020). It also expects to see more AI solutions in clinical practice based on evidence from clinical trials, with increasing focus on improved and scaled clinical decision-support (CDS). Whereas in another field, as manufacturing, AI-powered predictive maintenance allows for drastic reductions in costly unplanned downtime, as well as for extending the Remaining Useful Life (RUL) of production machines and equipment.

Alongside opportunities, AI brings also risks and challenges, as always occurs with new waves of technological innovation. It presents new and unknown risks that may or may not require ad hoc